

Pseudocode

① Write pseudocode to design a student class with attributes (name, rollno, marks) and Methods to :

- Accept student details
- Calculate average mark
- Display result (Pass/Fail)

Class Student

Private :

name

rollno

marks1, marks2, marks3

average

Public :

Method acceptDetails()

Input name

Input rollno

Input marks1, marks2, marks3

END Method

Method calculateAverage()

$$\text{average} = (\text{marks1} + \text{marks2} + \text{marks3}) / 3$$

End Method

Method displayResult()

Print "Name =", name

Print "Roll No =", rollNo

Print "Average =", average

IF average > 50 Then

Print "Result = PASS"

Else if

End Method

End Class

Main Program

Create objects s of Student

S. acceptDetails()

S. CalculateAverage()

S. displayResults()

END PROGRAM.

② Design a book class with private attributes (title, author, ISBN, isAvailable). Important methods to :

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods.

Class Book

Private :

title

author

ISBN

isAvailable

Public :

Method SetDetails (t, a, i)

title = t

author = a

ISBN = i

is Available = True

END Method

Method issueBook()

IF is Available = True then

is Available = False

Print "Book Issued"

Else

Print "Book Not Available"

End if

End Method

Method returnBook()

is Available = True

Print "Book Returned"

End Method

Method displayBook()

Print "Title =", title

Print "Author =", author

Print "ISBN =", ISBN

Print "Available =", is Available

End Method

End Class

Main Program

Create objects b1, b2, b3 of books

b1. SetDetails ("Java", "James", "101")

b2. SetDetails ("Python", "Guido", "102")

b3. SetDetails ("C++", "Bjarne", "103")

b1. IssueBook()

b1. displayBook()

b1. returnBook()

b1. displayBook()

End Program.

③ Write a pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation).

Implement methods to

- Calculate gross salary (basic + HRA)
- Calculate net salary after deductions
- Display pay slip.

= Class Employee

Private :

empId

name

basicPay

designation

grossSalary

netSalary

Public

Method acceptDetails()

Input empId

Input name

Input designation

Input basic Pay

End Method

Method calculateGrossSalary()

HRA = basicPay * 0.20

DA = basic Pay * 0.10

grossSalary = basic Pay + HRA + DA

End Method.

Method calculateNetSalary()

PF = basicPay * 0.12

Tax = grossSalary * 0.05

netsalary = grossSalary - PF - Tax

End Method.

Method displayPaySlip()

Print "Employee ID =", empId

Print "Name =", name

Print "Designation =", designation

Print "Basic Pay =", basic pay

Print "Gross Salary =", grossSalary

Print "Net Salary =", netsalary

End Method

End Class

Main Program

Create object e of Employee

e.acceptDetails()

e.calculateGrossSalary()

e.calculateNetSalary()

e.displayPaySlip()

End Program.

④ Create a Shape base class with an overridable calculateArea() method. Apply this to

- Circle - area using radius
- Rectangle - area using length and breadth
- Triangle - area using base and height

= class Shape

Method calculateArea()

Print "Area Calculation"

End Method

End Class

Class Circle Inherits Shape

Private:

radius

Method SetRadius (r)

radius = r

End Method

Method CalculateArea()

area = $3.14 * \text{radius} * \text{radius}$

Print "Circle Area =", area

End Method

End Class

Class Rectangle Inherits Shape

Private:

length, breadth

Method SetValues (l, b)

length = l

breadth = b

End Method

Method CalculateArea()

area = length * breadth

Print "Rectangle Area =", area

End Method

End Class

Class Triangle Inherits Shape

Private:

base, height

Method SetValues (b, h)

base = b

height = h

End Method

Method calculateArea()

Area = 0.5 * base * height

Print "Triangle Area -", area

End Method

End Class

Main Program

Create Objects c, r, t

c. Set Radius (5)

r. Set Values (8, 4)

t. Set values (10, 6)

c. calculateArea()

r. calculateArea()

t. calculateArea()

End Program

⑤ Design a Vehicle class with attributes (reg.No, owner, fuelType)

Extend it into

- TwoWheeler - with engine capacity
- Four Wheeler - with seating capacity and AC status.

≡ Class Vehicle

Protected:

regNo

owner

fuelType

Method SetDetails (r, o, f)

regNo = r

owner = o

fuelType = f

End Method

Method displayDetails()

Print regNo

Print owner

Print fuelType

End Method

End Class

Class Two Wheeler Inherits Vehicle

Private:

engineCapacity

Method SetEngine (cc)

engineCapacity = cc

End Method

End Class

Class Four Wheeler Inherits Vehicle

Private:

seatingCapacity

AC status

Method setFeatures (seat, ac)

seatingCapacity = seat

AC status = ac

End Method

⑥ Develop pseudocode for a Bank Account superclass and its Subclasses — SavingsAccount.

== Class BankAccount

Protected:

accNo

holder

balance

Method deposit (amount)

balance = balance + amount

End Method.

Method withdraw (amount)

End Method

End Class

Class SavingsAccount Inherits BankAccount

Method withdraw (amount)

IF amount $\leq$ balance Then

balance = balance - amount

Print "Withdrawal Successful"

Else

Print "Insufficient Balance"

End If

End Method

Method calculateInterest ()

interest = balance * 0.04

Print "Interest = ", interest

End Method

End class

class CurrentAccount Inherits BankAccount.

Private:

Overdraftlimit = 5000

Method withdraw (amount)

IF amount $\leq$ balance + overdraftlimit Then

balance = balance - amount

Print "Withdrawal Successfull with Overdraft"

Else

Print "Overdraft limit Exceeded"

End IF

End Method

End class.

⑦ Design a patient base class and analyse how it extends into:

- Inpatient

- Outpatient

= class Patient

Protected:

patientId

name

Private:

diagnosis

prescription

Method generate Bill()

End Method

End class

Class InPatient Inherits Patient

Private;

WardNo

admissionDate

daily charges

days.

Method generate Bill()

bill = daily charges * days

print "InPatient Bill =", bill

End Method.

End Class

Class OutPatient Inherits Patient

Private:

AppointmentDate

Consultation fee

Method generate Bill()

Print "OutPatient Bill =", ConsultationFee

End Method

End Class

system:

- Electronics - with warranty, brand, voltage
- Clothing - with size, fabric, gender
- Grocery - with expiry date and weight.

= Class Product

Protected:

product Name

price

Method applyDiscount()

End Method

End Class

Class Electronics Inherits Product

Method applyDiscount()

discount = price * 0.10

print "Final Price =", price - discount

End Method

End Class

Class Clothing Inherits Product

Method applyDiscount()

discount = price * 0.20

Print "Final Price =", price - discount

End Method

End Class

Class Grocery Inherits Products

Method applyDiscount()

discount = price * 0.05

Print "Final Price =", price - discount

End Method

End Class